StructuralCost:

a controlled reading time dataset for modeling human sentence processing difficulty

Nina Nusbaumer[1], Iria de-Dios-Flores[2], Corentin Bel[3,4], Christophe Pallier[3,5], Guillaume Wisniewski[1], Benoît Crabbé[1]

[1]Université Paris Cité, LLF, CNRS, France; [2]Universitat Pompeu Fabra, COLT, Spain; [3]Unicog, Neurospin, CEA, France; [4]LNC2, ENS-PSL, France; [5]INSERM, CNRS, France

Can LMs serve as models of human sentence processing?

CONTEXT

Comprehension relies on:

- **Prediction:** unexpected words cost more (surprisal)
- **Working memory:** longer dependencies cost more

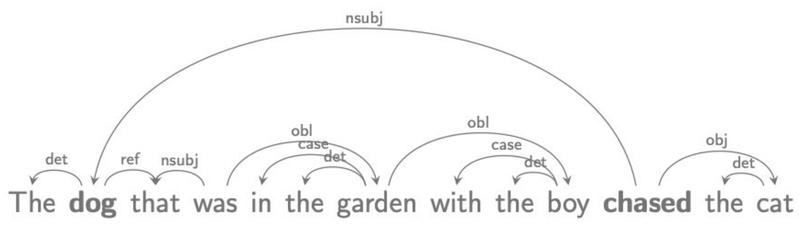
LMs are natural candidates: word probabilities (surprisal) correlate with reading times.

Expectation: **LMs underestimate structural difficulty**

- No memory decay,
- Huge context window

MOTIVATION

Theoretical consensus



Methodological challenge: large naturalistic reading corpora have **low control** while psycholinguistic datasets are **too small** for model evaluation.

STIMULI STRUCTURE

- Systematic **subject–verb dependency length** variation (0 / 4 / 9 words) and **embedding depth**
- Generated with **LLaMA-3.2** and **manually reviewed** for coherence, S-V predictability

Biases controlled for:
Antilocality: S-V relations kept plausible but not strongly expected • **Priming**: sentences presented in isolation • **Speeding**: all sentences constrained to 13–15 words

DATA COLLECTION

- Task: **Self-paced reading** (non-cumulative moving-window)
- Participants: **native English speakers** (Prolific)
- Working memory capacity measured via an **operation-span task**

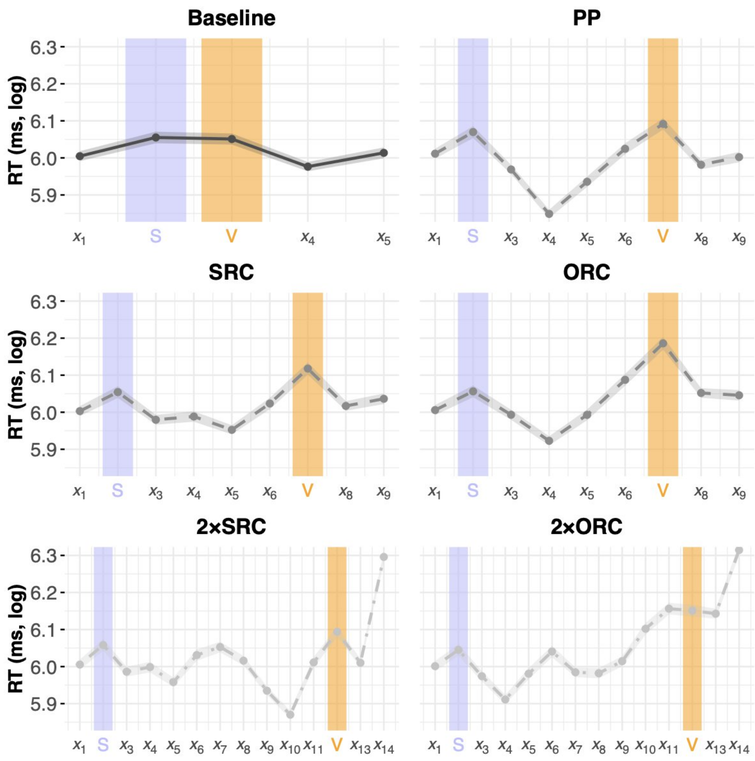
	Participants	Critical	Control
Per list	≈80	60	30
Total	475	27,103	13,663

StructuralCost
- ✓ Isolates the cost of S-V dependency length.
- ✓ 450 English sentences
- ✓ 475 participants
- ✓ ~ 40,800 observations

Condition	Dep.	Example Sentence
Critical sentences		
Baseline	0	The violonist leaves the stage before the audience loudly applauds the great performance.
PP	4	The violonist in the large orchestra leaves the stage before the audience loudly applauds.
SRC	4	The violonist that followed the conductor leaves the stage before the audience loudly applauds.
ORC	4	The violonist that the conductor followed leaves the stage before the audience loudly applauds.
2×SRC	9	The violonist that followed the conductor that worked in the orchestra leaves the stage.
2×ORC	9	The violonist that the conductor that the whole orchestra admired followed leaves the stage.
Control sentences		
Baseline	0	The waiter stacks the plates with the dirty apron next to the kitchen sink.
PP-short	4	The waiter with the dirty apron stacks the plates next to the kitchen sink.
PP-long	9	The waiter with the dirty apron next to the kitchen sink stacks the plates.

HUMAN BEHAVIOR

Does dependency length predict reading times?



Baseline model: log(V_RT) ~ word length + log(freq)
+ (1 | participant) + (1 | item) + (1 | order)

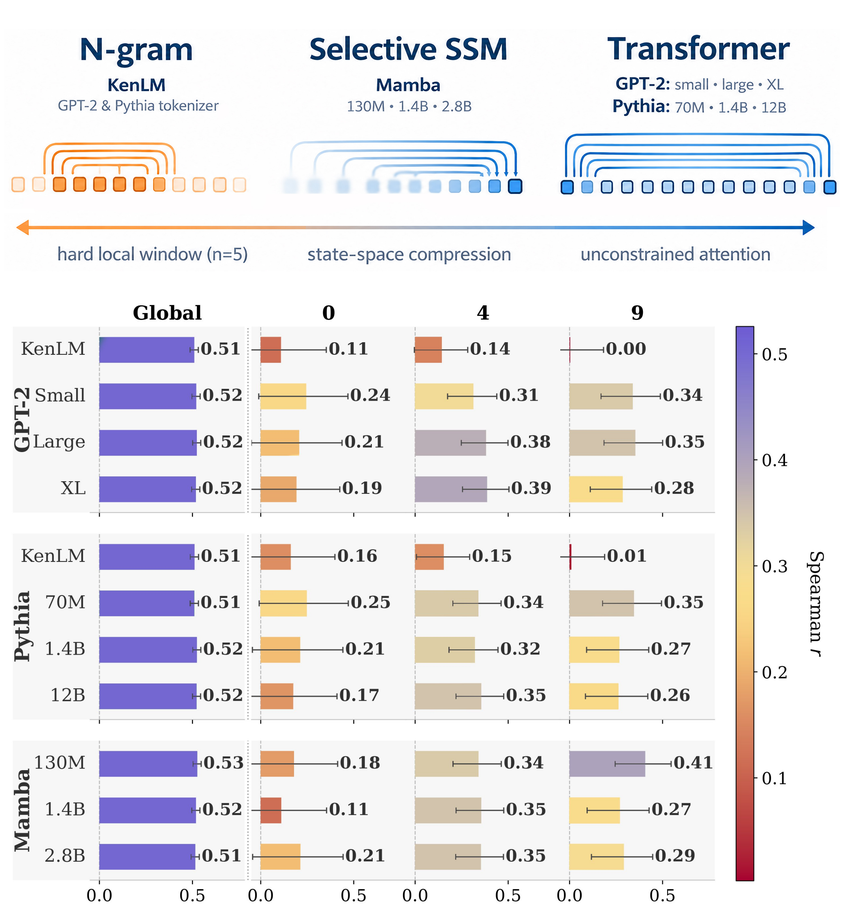
Distance model: log(V_RT) ~ **dep length** + word length + log(freq)
+ (1 | participant) + (1 | item) + (1 | order)

At the verb, humans pay for distance, but for structure too.

- ✓**Distance costs**: dependency length significantly improves model fit (χ^2= 21.6, p<0.001), independently of word length and frequency
- ✓**Structure costs on top of distance:** PP < SRC < ORC
- ✓**Costs don't stack with nesting depth**: double embedding is not harder than single (+14 / +63 ms vs. +49 / +93 ms)

LANGUAGE MODEL ALIGNMENT

Do LMs track the pattern?



LMs fit RTs globally, but fall short at the memory-sensitive position.

- ✓**Global fit** is real but **uninformative**
- ✓The **fit drops** where **memory matters**
- ✓**Surprisal doesn't capture the whole cost:** dependency length improves fit beyond surprisal (χ^2 = 5-11, p < .05)
- ✓**LMs get the order, not the size**: PP < SRC < ORC but predict ORC ≤ 2 ms
- ✓**Scale does not close the gap**

Do LMs predict the effect size?

Method: conversion factor (ms per bit) estimated on an independent corpus applied to StructuralCost.

Predicted effects are 1–3 orders of magnitude too small.

Condition	Observed (ms)	Predicted (ms) (best of 4 models)
PP	22	1.0
SRC	49	0.0
ORC	**93**	**2.0**
2×SRC	14	-0.2
2×ORC	63	1.7

Values: additional cost compared to the baseline.

Contributions

1. **A resource:** a large, controlled dataset of dependency length effects.
2. **A finding:** syntactic structure matters beyond linear distance.
3. **A diagnostic:** LMs fit reading times well overall but miss the structural difficulty.

Acknowledgement
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Paper

Dataset

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